Environmental Engineering

Unit II: Water Treatment

Syllabus

Objectives - Unit operations and processes - Principles, functions and design of water treatment plant units, aerators, flash mixers, coagulation and flocculation - Clariflocculator - Plate and tube settlers - Pulsator clarifier - Sand filters - Disinfection - Softening, removal of iron and manganese - Defluoridation - Desalination process - Residue management - Construction, operation and maintenance aspects.

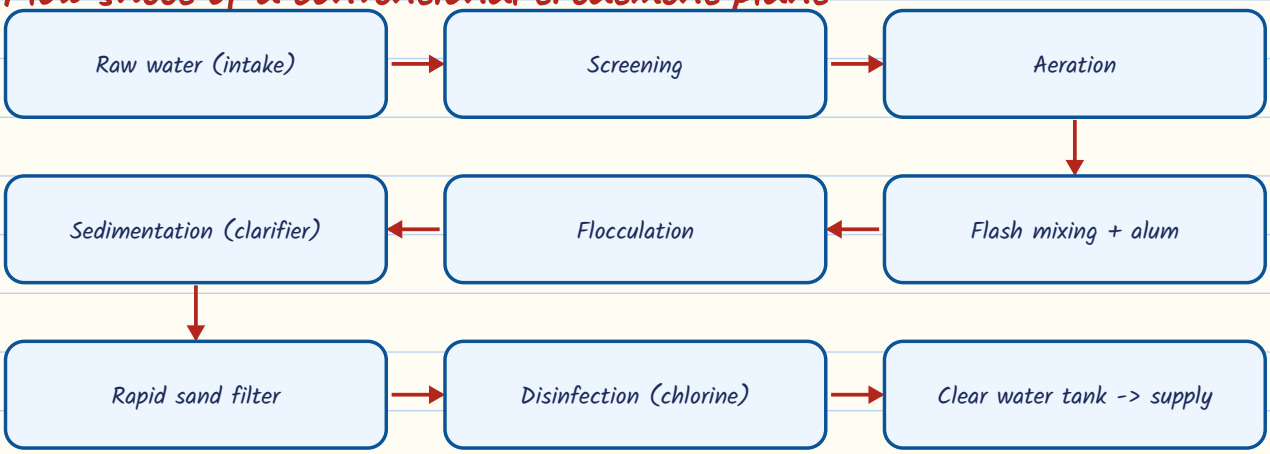
1. Objectives of Water Treatment

- To remove **suspended and colloidal impurities** (turbidity, colour)
- To remove **taste, odour** and dissolved gases
- To remove **pathogenic bacteria** and make water bacteriologically safe
- To reduce **hardness, iron, manganese, fluoride** and other harmful chemicals
- To make water **potable, palatable** and suitable for industries (non-corrosive, non-scale forming)

Unit operations and unit processes

- **Unit operations** - physical forces: screening, aeration, sedimentation, filtration, flotation
- **Unit processes** - chemical or biological reactions: coagulation, disinfection, softening, ion exchange, adsorption

Flow sheet of a conventional treatment plant



Choice of units depends on raw water quality. For a **clear ground water** only disinfection (and iron removal) may be enough; for a **turbid river water** the full flow sheet is needed.

2. Screening and Plain Sedimentation

- **Screens** are first unit; **coarse screens** (bar spacing 50-150 mm) remove floating debris, **fine screens** (6-12 mm) remove smaller matter; velocity through the screen below 0.8 m/s
- **Plain sedimentation** - settling of suspended particles under gravity without chemicals; removes sand and silt (above 0.02 mm), reduces load on later units
- Used as **pre-sedimentation** for very turbid rivers (turbidity above 500 NTU)

Theory of sedimentation

Stoke's law (settling velocity of a discrete particle)

$v_s = g (S_s - 1) d^2 / (18 nu)$

$g = 9.81 m/s^2$, S_s = specific gravity of particle, d = diameter, nu = kinematic viscosity

Valid for laminar flow, Reynolds number less than 1

- **Ideal settling basin zones:** inlet, settling, sludge, outlet
- **Surface overflow rate (SOR)** = $Q / A = v_0$; all particles with $v_s >= v_0$ are completely removed (independent of depth!)
- **Detention time** $t = V / Q$ = (volume of tank) / (flow rate)
- **Displacement efficiency** / flow-through period = length / velocity of flow

Design parameter	Value
Detention time (plain)	3 - 4 h
Detention time (with coagulation)	2 - 2.5 h
Surface overflow rate (plain)	15 - 30 $m^3/m^2/day$
SOR (with coagulant)	30 - 40 $m^3/m^2/day$
Horizontal velocity	Not more than 0.3 m/min (0.5 cm/s)
Depth	2.5 - 5 m
Length : Width	3:1 to 5:1 (rectangular); diameter up to 60 m (circular)
Weir loading	Below 300 $m^3/m/day$

Types of sedimentation tanks

- **Rectangular** - horizontal flow, length 3-5 times width, sludge scraped to a hopper
- **Circular** - radial flow, inlet at centre, outlet at periphery, rotating scraper
- **Upward flow tank** - used with sludge blanket
- **Continuous flow** vs **fill and draw** (intermittent) tanks

Design example

Q = 6 MLD, detention time 3 h. Volume = $6 \times 10^6/1000 \times 3/24 = 750\ m^3$.

Depth 3.5 m gives area = $214\ m^2$; take L : B = 4 : 1, B = 7.3 m, L = 29.3 m.

3. Aeration and Aerators

Aeration brings water and air into intimate contact to add oxygen and remove dissolved gases (CO_2, H_2S) and volatile compounds that cause taste and odour; it also oxidises **iron and manganese**.

Types of aerators

Aerator	Working
Gravity (cascade) aerator	Water falls over steps, trays or weirs; simple, no power
Spray aerator	Water sprayed through nozzles into air; 1.5-2 m head needed
Diffused air (bubble) aerator	Compressed air through porous plates at tank bottom
Mechanical aerator	Rotating paddles or turbines; surface agitation
Tray aerator	Perforated trays filled with coke or gravel to increase contact

- Contact time: 10-30 min in diffused aerators; air requirement 0.3 - 0.8 m^3 air per m^3 water
- The **flash mixer** is also called a rapid mixing unit (see next section)

4. Coagulation and Flocculation

Colloidal particles (0.001 - 1 micron) carry **negative charge** and repel each other, so they do not settle. **Coagulation** neutralises this charge using chemicals (coagulants), forming micro-flocs. **Flocculation** is slow mixing which joins micro-flocs into large settleable flocs.

Common coagulants

Coagulant	Formula	Notes
Alum (aluminium sulphate)	$Al_2(SO_4)_3.18H_2O$	Most common, cheap, pH 6.5-8.5, dose 10-50 mg/L
Ferrous sulphate (copperas)	$FeSO_4.7H_2O$	Needs lime; pH above 8.5
Ferric chloride	$FeCl_3$	Works at wide pH 4-11
Sodium aluminate	$NaAlO_2$	Used for coloured, soft waters
Poly aluminium chloride (PAC)	$Al_n(OH)_mCl$	Efficient, low sludge
Polyelectrolytes	Organic polymers	Coagulant aids, strengthen flocs

Reaction of alum with alkalinity

$Al_2(SO_4)_3 + 3Ca(HCO_3)_2 \rightarrow 2Al(OH)_3 + 3CaSO_4 + 6CO_2$

$Al(OH)_3$ is a gelatinous precipitate that traps colloids (**sweep floc**). If natural alkalinity is low, add lime or soda ash.

1 mg/L alum needs about 0.5 mg/L alkalinity as $CaCO_3$.

- **Optimum dose** found by **Jar test** - six beakers, different doses, rapid mix 1 min at 100 rpm, slow mix 20 min at 30 rpm, settle 30 min; the dose giving the clearest water is the optimum
- Factors: turbidity, pH, temperature, alkalinity, mixing intensity
- **Coagulant aids:** lime, activated silica, bentonite clay, polymers

Flash (rapid) mixer

- Provides violent mixing to disperse coagulant quickly: detention time **30 - 60 seconds**; velocity gradient $G = 300 - 1000\ s^{-1}$
- Types: **hydraulic** (weirs, hydraulic jump, Parshall flume, baffles), **mechanical** (propeller or turbine mixers)
- Mixing power: $P = G^2\ mu\ V$ (mu = dynamic viscosity, V = volume)

Flocculator

- Gentle stirring, $G = 10 - 75\ s^{-1}$, detention time **20 - 40 min**; Gt = 20,000 - 1,00,000 (dimensionless)
- Types: **mechanical** (paddle flocculators, slow-speed paddles with peripheral velocity 0.2-0.6 m/s) and **hydraulic** (baffle-wall: horizontal / over-and-under baffles with velocity 0.15-0.3 m/s)

- Water velocity must stay below 0.3 m/s in the outlet channel so that floc does not break



5. Clariflocculator

A **clariflocculator** combines the flocculation and sedimentation in one circular structure: a central flocculation zone surrounded by an outer clarification zone. It saves space and cost.

- Water with coagulant enters the **inner circular flocculation chamber** (mechanical paddle stirrers, detention 20-40 min)
- Flows under a skirt wall to the **outer annular clarifier** with slow upward/radial flow
- Settled sludge scraped to the central sump by a rotating bridge scraper
- Clarified water collected by peripheral launder over weirs
- Typical: flocculator detention 20-30 min, clarifier 2 h, SOR 30-40 $m^3/m^2/day$
- Used for large plants, with high turbidity raw water

6. Plate and Tube Settlers

These are **high rate settlers**: inclined plates or tubes are placed inside the settling basin to increase the **effective settling area** without increasing the tank area. Settling distance becomes very small (5 cm) so particles settle quickly.

- **Tube settlers** - square, hexagonal or circular tubes 25-50 mm diameter, inclined at **60 degrees** to horizontal; flow upward; sludge slides down the tubes by gravity
- **Plate settlers (lamella)** - parallel plates 50-100 mm apart, at 55-60 degrees; plates of PVC or FRP
- Reduce detention time from 2-3 h to **15-30 min**; loading rate up to 3-4 times conventional
- Used to upgrade overloaded old plants

Advantages and limits

* Compact, 40-70% area saving; good removal of fine flocs
* Needs good pre-flocculation, periodic cleaning (algae, sludge in tubes)
* Not for water with very high variations in turbidity

7. Pulsator Clarifier

A **sludge blanket (solids-contact)** clarifier with pulsating inflow. Raw water with coagulant is fed into a vacuum chamber; air is periodically released so that water is **pulsed** into the tank through perforated laterals at the bottom.

- Pulses keep the **sludge blanket** uniformly suspended and homogeneous (flocs formed in the blanket), providing very effective clarification
- Blanket height 1.5-2 m; excess sludge flows to concentrators
- Flat-bottom, square or rectangular; rise rate 1.5 - 3 m/h
- Good turbidity removal, but sensitive to sudden changes of flow, requires skilled operation

8. Filtration

Filtration removes remaining suspended and colloidal matter, bacteria and flocs when water passes through a porous bed (sand). It removes **90-99% of bacteria**, and cuts turbidity to below 1 NTU.

Feature	Slow sand filter	Rapid sand filter
Rate of filtration	100 - 200 L/h/m^2 (0.1-0.4 m/h)	3000 - 6000 L/h/m^2 (3-6 m/h)
Area required	Large	Small (1/30)
Sand effective size	0.2 - 0.4 mm, uniformity coeff. 2-3	0.45 - 0.7 mm, UC 1.3-1.7
Depth of sand	90 - 110 cm	60 - 75 cm
Pre-treatment	Plain sedimentation only	Coagulation + sedimentation needed
Mechanism	Biological action (schmutzdecke)	Mainly physical straining, adsorption
Cleaning	Scraping top 1-2 cm every 1-3 months	**Backwashing** every 24-48 h
Bacteria removal	98-99%	80-90%, so disinfection needed
Cost	High initial, low running	Low initial, high running

Rapid gravity filter - components

- **Filter bed** - 60-75 cm sand above 45 cm gravel in graded layers
- **Under-drainage system** - perforated laterals and manifold (central main), collects filtered water, distributes backwash
- **Filter box** - 3 to 4 m deep, RCC; area 10-100 m^2 each; at least 2 units
- **Wash-water troughs** above the bed
- **Rate controller** keeps rate constant, **loss of head gauge**
- **Backwashing** - reverse flow of clean water at 0.6 - 1 m/min to expand bed by 30-50%, for 5-10 min; used wash-water is 1-3% of filtered water; **air scour** helps
- Filter is washed when head loss reaches 2.5 - 3 m or effluent turbidity increases
- **Operational troubles:** air binding (negative pressure), mud balls, cracking of the bed, sand boils, filter clogging

Other filters

- **Pressure filters** - closed steel cylinders, rate 6-15 m/h, used in small plants, swimming pools, industry
- **Dual-media filters** - anthracite + sand; allow deeper penetration and higher rates
- **Multi-media filters, membrane filters (UF, MF), activated carbon filters**
- **Gravity filters designed:** Area = Q / rate; e.g., 6 MLD at 5 m/h needs 50 m^2

9. Disinfection

Disinfection means the **killing of pathogenic organisms** (not total sterilisation). It is the last treatment barrier. Ideal disinfectant: kills bacteria in short time, harmless to humans, leaves a **residual**, cheap, easy to apply and measure.

Methods

- **Chlorination** - most common; chlorine gas, bleaching powder (30-35% available Cl_2), sodium hypochlorite, chlorine dioxide
- **Ozonation** - strong, no taste, no residual; costly
- **Ultraviolet (UV) radiation** - 254 nm, no residual, needs clear water
- **Boiling** (household), **iodine, silver, potassium permanganate**

Chemistry of chlorination

$Cl_2 + H_2O \rightarrow HOCl + HCl$; $HOCl \leftrightarrow H^+ + OCl^-$

HOCl (**hypochlorous acid**) is the strong germicide (about 80 times more effective than OCl^-); it dominates at **pH below 7.5**

Free residual chlorine = $HOCl + OCl^-$; combined = chloramines (NH_2Cl, $NHCl_2$)

Chlorine demand, dose, residual

- **Chlorine demand** = dose applied - residual after contact time
- **Break-point chlorination** - chlorine is added until all ammonia and organics are oxidised; beyond the break point, the residual is **free** chlorine; ensures complete disinfection and removes taste and odour
- **Contact time** 30 min minimum, **residual chlorine** 0.2 mg/L at the consumer end (0.5 at the plant outlet)
- Dose of chlorine 0.2 - 0.5 mg/L for filtered water; bleaching powder dose = (chlorine dose x water volume) / (% available Cl_2)

Type of chlorination	Meaning
Plain / post-chlorination	Chlorine after filtration
Pre-chlorination	Before sedimentation, to oxidise and control algae
Super-chlorination	Large dose, then dechlorination with SO_2 or sodium thiosulphate
Dechlorination	Removing excess chlorine
Break-point chlorination	Dose beyond the break point

Disadvantage: formation of **trihalomethanes (THMs)**, which are carcinogenic, when chlorine reacts with organic matter. Chlorine applied by **chlorinators** (solution feed, vacuum type).

10. Water Softening

Hardness causes scale formation in boilers, wastage of soap, and is unsuitable for laundry and industry. Softening removes Ca and Mg ions. Needed when hardness exceeds 200 mg/L (up to 100 mg/L for industry).

Method	Principle
Lime-soda process	Lime removes carbonate hardness (temporary) and Mg; soda ash removes non-carbonate (permanent) hardness. Precipitates $CaCO_3$ and $Mg(OH)_2$
Zeolite (ion exchange) process	Water passes through sodium zeolite; Ca and Mg exchanged for Na. Regenerated with brine (NaCl)
Boiling	Removes temporary hardness only
Clark's method	Lime added to temporary hardness
Demineralisation	Cation and anion resin exchange, gives ion-free water

Method	Principle
Reverse osmosis / electrodialysis	Membrane separation

Lime-soda reactions

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$

$MgSO_4 + Ca(OH)_2 \rightarrow Mg(OH)_2 + CaSO_4$; $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 + Na_2SO_4$

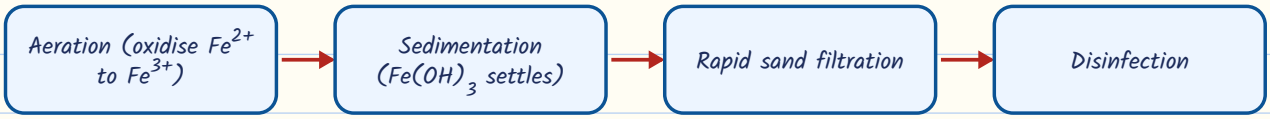
Zeolite: $Na_2Z + Ca(HCO_3)_2 \rightarrow CaZ + 2NaHCO_3$

Lime-soda vs zeolite

* Lime-soda: handles large hardness, low cost chemicals, sludge produced, residual hardness 30-50 mg/L
* Zeolite: no sludge, residual hardness near zero (0-5 mg/L), but only for clear water with low iron and turbidity, brine for regeneration

11. Removal of Iron and Manganese

- Present in ground water as dissolved **ferrous (Fe^{2+})** and **manganous (Mn^{2+})** forms with CO_2 and bicarbonate
- Problems: red-brown stains on clothes and fixtures, metallic taste, iron bacteria, clogging of pipes; limit Fe 0.3 mg/L, Mn 0.1 mg/L



- **Oxidation methods:** aeration, chlorine, potassium permanganate (**best for Mn**, since Mn needs pH above 9 for oxidation by air)
- **Lime softening** raises pH and precipitates Fe and Mn
- **Manganese zeolite / greensand filter** (oxidising bed regenerated with $KMnO_4$)
- **Ion exchange** - for low concentrations

12. Defluoridation

Fluoride above **1.5 mg/L** causes **dental and skeletal fluorosis** (mottled teeth, bone deformity). Common in ground water of Andhra Pradesh, Rajasthan, Gujarat, Tamil Nadu. Defluoridation brings the fluoride within 1 mg/L.

Method	Details
Nalgonda technique	Developed by NEERI. Alum + lime (or sodium aluminate) added, flash mixing, flocculation, settling, filtration. Domestic (bucket) and community plants. Alum dose 350-700 mg/L
Activated alumina adsorption	Water passes through activated alumina bed; regeneration with alum or NaOH
Bone charcoal / activated carbon	Adsorption, low cost
Ion exchange resin	Anion resin
Reverse osmosis, electrodialysis	High efficiency, costly
Other	Mixing with low-fluoride water, rainwater harvesting

13. Desalination

Removal of dissolved salts from **brackish** (TDS 1000-10,000 mg/L) or **sea water** (TDS about 35,000 mg/L) to make it fresh (TDS below 500).

Process	Principle
Reverse osmosis (RO)	Pressure (above osmotic pressure, 15-70 bar) pushes water through a semi-permeable membrane against natural osmotic flow; salt rejection 95-99%; energy 3-5 kWh/m^3
Electrodialysis (ED)	Ions move through ion-selective membranes under an electric field; for brackish water
Distillation (multi-stage flash, MED)	Water evaporated and condensed; heat energy; sea water
Solar still	Sun heat; small scale
Ion exchange	Resins swap ions; low salinity
Freezing	Ice crystals pure; less popular

- RO needs **pre-treatment** (filtration, anti-scalants, pH control) to prevent membrane fouling; waste **reject (brine)** 20-50% must be disposed safely
- **Nano-filtration** - removes hardness and some salts at lower pressure

14. Residue (Sludge) Management

Treatment plants produce sludge from sedimentation tanks, wash water from filters, and brine / reject from softening and RO. These must be handled carefully to avoid pollution.

- **Sources:** alum sludge ($Al(OH)_3$), lime sludge ($CaCO_3$), backwash water, iron sludge
- **Thickening** (gravity thickener) -> **conditioning** (polymers) -> **dewatering** (sand drying beds, filter press, centrifuge, belt press) -> **disposal**
- Lime sludge can be **recalcined** and reused, or used in soil conditioning
- Filter backwash is **recycled** to the head of the plant after equalisation
- Disposal: landfill, lagoons, discharge to sewer, brick and cement making, **RO reject** to evaporation ponds or marine outfall

15. Construction, Operation and Maintenance

- **Construction:** treatment units in RCC, water-tight, placed on hydraulic grade line so that flow is by gravity, with bypass lines and flow measurement (Parshall flume, venturi)
- **Operation:** daily record of raw and treated water quality, chemical doses by jar test, chlorine residual testing, flow rates, head loss
- **Maintenance:** cleaning of sedimentation tanks (desludging), backwashing of filters, painting, lubrication of scrapers and mixers, calibration of chlorinators and meters
- Standby units for pumps and chemical feeders; trained operators and safety (chlorine leak kits, gas masks)
- Safety: chlorine is toxic - store cylinders upright, in cool, ventilated rooms; leaks detected using ammonia swab

Remember for exam

* Alum dose 10-50 mg/L; flash mixer 30-60 s; flocculation 20-40 min; sedimentation 2-3 h
* RSF rate 3-6 m/h, SSF 0.1-0.4 m/h; backwash every 24-48 h
* Chlorine residual 0.2 mg/L, contact time 30 min; HOCl best at low pH
* Zeolite = no sludge; lime-soda = sludge; Nalgonda uses alum + lime
* RO is the principal desalination process

Important questions

1. Draw a flow sheet of a conventional treatment plant and explain the function of each unit.
2. Explain the theory of sedimentation and the design of a rectangular settling tank.
3. What is coagulation? Explain coagulants, jar test, flash mixing and flocculation.
4. Describe the working of a clariflocculator and tube settlers.

5. Compare slow and rapid sand filters. Explain the backwashing process.
6. Explain disinfection by chlorination and break-point chlorination.
7. Describe softening by lime-soda and zeolite processes.
8. Explain the removal of iron, manganese and fluoride from water.
9. Discuss desalination processes and management of residues from water treatment.